

**Econ 630, Spring 2013:  
Final Exam**

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Name: \_\_\_\_\_

The exam consists of 3 questions. Your answer should be clearly labeled and legible. Show all your work. No credit will be given for answers that are not justified.

1. (30 pts) Consider the following simultaneous move game between two players:

|   | L           | M         | R         |
|---|-------------|-----------|-----------|
| U | -10    4    | 6       0 | -1    -1  |
| C | 0       6   | -1    -1  | -1    1   |
| D | 4       -10 | -1    -1  | 2       2 |

Figure 1

- (a) (10 pts) Find all rationalizable strategies.

**Solution:**  $C$  is strictly dominated by a mixing between  $U$  and  $D$ ,  $M$  is strictly dominated by a mixing between  $L$  and  $R$ . In the remaining  $2 \times 2$  game,  $U$  is strictly dominated by  $D$ . Therefore, the only set of rationalizable strategies is  $(D, R)$ .

(b) (5 pts) Find all Nash equilibria of this game.

**Solution:**  $(D, R)$ .

- (c) (5 pts) Consider a repeated game  $G^T$  that consists of repeating the stage game  $T$  times. Find the subgame perfect equilibria of  $G^T$ .

**Solution:** The unique equilibrium of the stage game will emerge as a unique equilibrium of the finitely repeated game.

- (d) (10 pts) Consider a repeated game  $G^\infty$  that consists of repeating the stage game infinite number of times with discount factor  $\delta$  for both players. Can you find an equilibrium that is preferred to playing a Nash equilibrium of the stage game infinite number of times? If so, explain the supporting trigger strategies carefully.

**Solution:** Consider the following strategy for player 1 and 2:

- Player 1: Play  $U$  in even periods and  $C$  in odd periods as long as player 2 has played  $M$  in even periods and  $L$  in odd periods in all prior periods. Otherwise, play  $D$ .
- Player 2: Play  $M$  in even periods and  $L$  in odd periods as long as player 1 has played  $U$  in even periods and  $C$  in odd periods in all prior periods. Otherwise, play  $R$ .

We want to show that such trigger strategies can be supported as an SPNE in the infinitely repeated game. Player 1, has no incentive to deviate from the trigger strategy in even periods, but she might have incentives to deviate in odd periods. No deviation incentives requires:

$$0 + 6\delta + 6\delta^3 + \dots \geq 4 + \frac{2\delta}{1 - \delta}$$

$$\delta \geq \sqrt{3} - 1$$

2. (40 pts) Suppose that an individual seller possesses a single indivisible object that she values at zero and wishes to sell. There are  $n$  risk neutral individuals (the buyers) interested in purchasing the object. Each buyer's valuation  $v_i$  is an iid draw from a uniform distribution on  $[0, 1]$ . Suppose that each buyer privately observes her valuation only after inspecting the object. Suppose that the seller posts an entry fee  $\psi$  and announces that any buyer paying this fee will be allowed to inspect the object and participate in a first price sealed bid auction with no reserve price. Bidders will observe how many other bidders have paid the entry fee before submitting their bid.

- (a) (5 pts) What is the ex-ante social surplus in this setting?

**Solution:** The ex-ante social surplus is simply the expected value of the first order statistic.

$$E(v_i^{max}) = \int_0^1 ny^n dy = \frac{n}{n+1}$$

- (b) (6 pts) Consider the subgame, in which  $M$  bidders have entered the auction. Find the symmetric Bayesian Nash equilibrium bidding strategy of the participants in the auction.

**Solution:** In a first price auction, each bidder will bid the expected value of the second highest valuation. Therefore,

$$b(v) = \frac{\int_0^v (M-1)y^{M-1} dy}{v^{M-1}} = \frac{M-1}{M}v$$

- (c) (6 pts) What is the expected equilibrium payoff of a bidder who learns that his valuation is  $v$  and observes that  $M$  bidders have entered the auction?

**Solution:**

$$\pi(v) = \left(v - \frac{M-1}{M}v\right)v^{M-1} = \frac{v^M}{M}$$

- (d) (6 pts) What is the ex-ante equilibrium payoff of a bidder from entering the auction if he is expecting  $M - 1$  other bidders to enter the auction.

**Solution:**

$$E\pi = \int_0^1 \frac{v^M}{M} dv = \frac{1}{M(M+1)}$$

- (e) (6 pts) What entry fee  $\psi(M)$  should the auctioneer set in order to induce entry by exactly  $M$  bidders in the auction? Calculate the expected equilibrium revenue by the seller when she sets the entry fee to  $\psi(M)$ .

**Solution:** In equilibrium, the highest entry fee that will induce entry by  $M$  bidders should satisfy  $\frac{1}{M(M+1)} = \psi$ . The seller's expected equilibrium revenue by the seller is:

$$E(R) = M\psi(M) + \int_0^1 M(M-1)(1-y)y^{M-1}dy = \frac{M}{M+1}.$$

Note that the second term is simply the expected value of the second order statistic.

- (f) (6 pts) Find the optimal entry fee  $\psi^*$  by the seller.

**Solution:** Since the seller's expected revenue is increasing in  $M$ , the seller would want to incentivize entry by all buyers. Therefore,  $\psi^* = \frac{1}{n(n+1)}$ .

- (g) (5 pts) Is the mechanism with an entry fee  $\psi^*$  optimal for the seller? Is it ex-ante efficient? Is it ex-post efficient?

**Solution:** The mechanism is both ex-ante and ex-post efficient since all bidders enter the auction and the item is allocated to the bidder who values it the most. It is also optimal for the seller since she extracts all the available surplus.

3. (30 pts) Consider the signalling game depicted by figure 2 with the following timing: 1) Nature draws the type of player 1 from a Bernoulli distribution with  $Pr(t_a) = 0.8$ . 2) Player 1 observes Nature's move and chooses one of two actions-  $L, R$ . 3) Player 2 observes player 1's action, but not his type and chooses one of two actions-  $U, D$ .

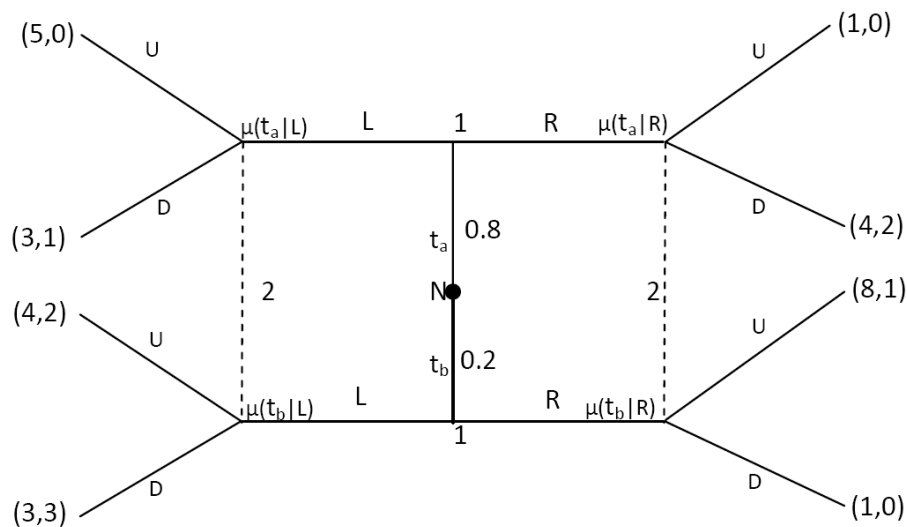


Figure 2

- (a) (10 pts) Find a separating PBE. Carefully specify the belief structure that supports this equilibrium.

**Solution:** The following is a separating equilibrium:  $(s_1(t_a) = R, s_1(t_b) = L, s_2(R|\mu^*(t_a|R)) = D), s_2(L|\mu^*(t_a|L)) = D, \mu^*(t_a|R) = 1, \mu^*(t_a|L) = 0)$ .

- (b) (10 pts) Is there a pooling PBE? If so, specify equilibrium strategies and a belief structure that supports such an equilibrium.

**Solution:** There is no pooling equilibrium. If the two pool on  $L$ , player 2 will play  $D$ , resulting in a payoff of 3 for both types. It is straightforward to verify that for whatever action (pure or mixed) by player 2 at information set reached when  $R$  is played, at least one of the types will have incentives to deviate to  $R$ . If the two pool on  $R$ , the second player would play  $D$ , resulting in a payoff of 1 for  $t_b$ . Therefore,  $t_b$  will have incentives to deviate to  $L$  since it results in a payoff of 3 independent of the receiver's belief at the off-equilibrium info set.

- (c) (10 pts) Is there a PBE, in which type  $t_a$  plays  $R$  and  $L$  with positive probability? If so, specify equilibrium strategies and a belief structure that supports such an equilibrium.

**Solution:** Let  $\alpha(t)$  denote the equilibrium probability of type  $t$  playing  $L$  in equilibrium and  $\beta(a_1)$  be the equilibrium probability of player 2 playing  $U$  upon observing  $a_1$ . The following constitutes a PBE:

$$\alpha(t_a) = \frac{7}{8}, \alpha(t_b) = 0, \beta(R) = \frac{1}{3}, \beta(L) = 0, \mu^*(t_a|R) = \frac{1}{3}, \mu^*(t_a|L) = 1$$